

Original Article

Sidaxue ameliorates rheumatoid arthritis by inhibiting synovial neutrophil migration: Insights from an integrated multi-omics analysis

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ABSTRACT

Background: The therapeutic outcomes for rheumatoid arthritis (RA) remain unsatisfactory. Neutrophil migration and activation in the synovium are pivotal drivers of RA pathology. Sidaxue (SX), a traditional Miao herbal formula, has shown anti-RA potential, yet its mechanism concerning neutrophil migration is unclear.

Purpose: This study aims to evaluate the effects of SX on neutrophil migration in RA and to uncover the novel molecular mechanisms.

Methods: We integrated bulk RNA sequencing (RNA-seq) and single-cell RNA-seq (scRNA-seq) with Mendelian randomization (MR) to pinpoint critical targets. Ultra-high-performance liquid chromatography-mass spectrometry (UHPLC-MS) characterized SX's components, with binding affinities evaluated by molecular docking and dynamics simulations. Functional assays used SX-containing serum *in vitro* and a collagen-induced arthritis (CIA) model *in vivo*.

Results: Multi-omics analysis identified the RAC1/PAK1 pathway as central to neutrophil migration in RA. MR analysis nominated PAK1 as a potential drug target. We identified 38 major compounds in SX, which demonstrated high binding affinity to RAC1 and PAK1. SX-containing serum potently inhibited phorbol 12-myristate 13-acetate (PMA)-induced neutrophil migration, neutrophil extracellular traps (NETs) formation, and pro-inflammatory cytokine release. These effects were mediated through suppression of the RAC1/PAK1/LIMK1/Cofilin1 signaling axis, a finding corroborated by rescue experiments with the modulator Fingolimod hydrochloride (FTY). *In vivo*, SX alleviated joint swelling and synovitis in CIA rats, concurrently downregulating this pathway.

Conclusion: SX ameliorates RA by inhibiting neutrophil migration and activation via the RAC1/PAK1/LIMK1/Cofilin1 pathway. This work unveils the immune-regulatory mechanism of SX, supporting its development as a natural therapeutic for RA.

Introduction

Rheumatoid arthritis (RA) is a systemic autoimmune disorder characterized by chronic synovitis and destruction of articular cartilage and bone. Its pathogenesis remains incompletely understood, though current evidence suggests a complex network involving genetic susceptibility, epigenetic regulation, environmental triggers, and disruption of immune homeostasis (Chen et al., 2024; D'Orazio et al., 2024; Misra, 2025). The global incidence of RA is rising, with current estimates ranging from approximately 0.24% to 1% worldwide (Xiaochen et al., 2025). Although recent studies have progressively identified novel

therapeutic targets for RA (Chen et al., 2023b, 2023a, 2021; Lohmander et al., 2014), effective clinical treatments remain limited. Therefore, exploring immunotherapeutic strategies for RA holds significant scientific and clinical value for elucidating its inflammatory mechanisms and improving treatment outcomes.

Neutrophils play a positive role in immune responses and tissue regeneration. However, in RA, they are considered key factors in promoting inflammation and pathological damage (Bissenova et al., 2022). Neutrophils are typically round and small with large, lobulated or horseshoe-shaped nuclei (McKenna et al., 2021). Neutrophils are abundantly and persistently present in the synovial fluid of RA patients

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(O’Neil and Kaplan, 2019). Under chemotactic guidance, neutrophils highly express integrin CD11b/CD18, which binds to ICAM-1 on joint microvascular endothelial cells, facilitating transendothelial migration (Wang et al., 2025). Once within the synovium, neutrophils interact with fibroblast-like synoviocytes (FLSs), exacerbating local inflammation and releasing citrullinated antigens via neutrophil extracellular traps (NETs), thereby breaking immune tolerance (Deng et al., 2025). Neutrophils migrate to the site of inflammation, abnormally accumulate, and release ROS and matrix metalloproteinases (MMP-8 and MMP-9), which degrade the cartilage surface and promote the breakdown of type II collagen in articular cartilage (Li and Lu, 2024). Therefore, neutrophils are not only early participants in RA inflammation but also key drivers of chronicity and sustained autoimmunity. Inhibiting neutrophil migration may represent a potential therapeutic strategy for improving inflammatory responses and tissue damage in RA.

Non-steroidal anti-inflammatory drugs and glucocorticoids are first-line therapies for RA. While effective in alleviating clinical symptoms and signs of inflammation, their use is associated with significant adverse effects, including gastrointestinal and renal damage (Baoqi et al., 2021; Delteil et al., 2023; Takanashi et al., 2025). Biologics such as TNF- α antagonists, IL-1 inhibitors, B-cell depleting agents, and T-cell blockers offer substantial efficacy but increase the risk of immune tolerance issues, infections, and malignancies (Burmester and Pope, 2017; Mitsuboshi, 2021). Recently, traditional Chinese medicine (TCM) has demonstrated considerable potential in RA treatment (Liao et al., 2025). The discovery of highly effective and low-toxicity natural compounds remains a major focus in modern drug development (Huang et al., 2025). Sidaxue (SX), a TCM formulation composed of *Spatholobus suberectus* Dunn (Jixueteng), *Sargentodoxa cuneata* (Oliv.) Rehd. et Wils. (Daxueteng), *Toddalia asiatica* L. Lam. (Jianxuefei), and *Periploca forrestii* Schltr (Heiguteng), has shown anti-rheumatoid effects in preliminary studies, though its precise mechanisms and active components require further elucidation (Wu et al., 2022).

Here, we employed an integrated multi-omics framework. We combined bulk RNA sequencing (RNA-seq) analysis, weighted gene co-expression network analysis (WGCNA), single-cell RNA sequencing (scRNA-seq) analysis, and Mendelian randomization (MR) to identify genes associated with neutrophil migration in RA synovial tissue. We then used Ultra-high-performance liquid chromatography–mass spectrometry (UHPLC-MS) to characterize the major compounds in SX, and performed molecular docking and dynamics simulations to evaluate the binding affinity of these compounds to neutrophil migration-related genes. Finally, through *in vitro* and *in vivo* experiments, we validated the role and mechanism of SX in regulating neutrophil migration, thereby providing new insights into natural product-based therapies for RA.

Materials and methods

Bioinformation analysis

Data source

Genes associated with neutrophil migration were obtained from the GeneCards database (<https://www.genecards.org/>). To ensure the quality of the resulting gene set, only those genes with a relevance score greater than 8 were selected. Bulk RNA-seq data from RA synovial tissues were obtained from the GEO database (accession numbers: GSE1919, GSE55235, and GSE77298), comprising a total of 31 cases and 22 healthy controls. scRNA-seq data were derived from the GEO database under accession number GSE221704. The dataset comprises femoral bone marrow tissues from three DBA/1J mice: one with collagen-induced arthritis (CIA), one vehicle-treated control, and one CIA mouse treated with soyasaponin Bb. GWAS data for RA (ID: ebi-a-GCST90018873) were sourced from IEU OpenGWAS (<https://gwas.mrcieu.ac.uk/>), including 216 cases and 409,001 controls. eQTL data were also acquired from IEU OpenGWAS under the identifier eqtl-a-

ENSG00000149269 (PAK1). Data sources and details are summarized in Table 1.

Bulk RNA-seq analysis

Batch effects across samples were removed using the sva package (v3.52.0) in R. Differentially expressed genes (DEGs) were identified using the limma package (v3.60.6) based on the following criteria: $|\log_2FC| > 0.5$ and adjusted P -value < 0.05 after applying the Benjamini-Hochberg (BH) correction. A weighted gene co-expression network was constructed using the WGCNA package (v1.73) to identify modules highly associated with RA (Langfelder and Horvath, 2008). Sample data were preprocessed and outliers were removed prior to constructing the correlation matrix. The optimal soft-thresholding power was selected to cluster the adjacency matrix and identify key modules. The module showing the strongest positive correlation with the RA disease group was selected for further analysis. Gene significance (GS) for RA and module membership (MM) within the hub module were calculated for each gene. The overlapping genes among DEGs, module genes, and neutrophil migration-related genes were identified using the ggvenn package (v0.1.10). Kyoto Encyclopedia of Genes and Genomes (KEGG) and Gene Ontology (GO) analyses were performed using the clusterProfiler package (v4.12.6), with an adjusted P -value < 0.05 set as the significance threshold.

scRNA-seq analysis

Single-cell transcriptomic analysis was employed to evaluate the druggability potential and cell-type-specific expression patterns of candidate genes in RA. The Seurat package (v5.2.1) was used to process the scRNA-seq data. Cells were retained only when they expressed ≥ 200 genes, were detected in ≥ 3 cells, exhibited $< 3\%$ hemoglobin reads, and contained $< 10\%$ mitochondrial transcripts. The top 2000 most variable genes were selected for principal component analysis (PCA). Dimensionality reduction and visualization were performed using UMAP based on the first 20 principal components. Cell clusters were manually annotated using well-established markers from previous studies (Ding et al., 2024) and the CellMarker 2.0 database (Hu et al., 2023) to ensure accurate cell type identification. Gene expression correlations were assessed using Pearson correlation method.

MR analysis

Extract SNPs strongly correlated within the 100-kb range upstream and downstream of cis-eQTLs ($P < 5E-08$), and filter them with an effect allele frequency > 0.01 . To minimize the impact of linkage disequilibrium (LD) and ensure the independence of instrumental variables (IVs), a clumping threshold ($R^2 < 0.001$ within a 10,000kb distance) was applied to filter the SNPs. The F -statistic [$F = (N - k^{-1})/k(R^2/(1 - R^2))$] was calculated, and IVs with an F -statistic less than 10 were excluded as they were considered weak instrumental variables. The genetic instrumental variables for MR analysis were required to meet three MR assumptions: (1) SNPs were directly associated with the exposure; (2) SNPs were not related to the exposure-outcome confounders; (3) SNPs

Table 1
Data information utilized in the research.

Accession numbers	Source	Sample information
Neutrophil migration	GeneCards database	751 neutrophil migration-related genes
GSE1919	GEO database	5 HC and 5 RA samples
GSE55235	GEO database	10 HC and 10 RA samples
GSE77298	GEO database	7 HC and 16 RA samples
GSE221704	GEO database	1 control, 1 RA and 1 treatment samples from bone marrow of femur
ebi-a-GCST90018873	IEU OpenGWAS	216 RA cases and 409,001 controls
eqtl-a-ENSG00000149269	IEU OpenGWAS	31,684 samples from whole blood

affected the outcome only through the exposure (Davies et al., 2018). Inverse variance weighting (IVW), Simple Median, Weighted Median, Simple mode and Weighted Mode were used to estimate causality. If the *P*-value of the results obtained by the IVW method was less than 0.05, it was regarded as having a causal relationship. Pleiotropy was evaluated by the MR-Egger test, and potential heterogeneity was measured by Cochran's *Q* test with IVW and MR-Egger. *P* > 0.05 indicated the absence of both heterogeneity and pleiotropy. Meanwhile, the Steiger filtering method was employed to exclude potential reverse causality.

UHPLC-MS analysis

SX extracts were analyzed using a Vanquish UHPLC system (Thermo Scientific, Waltham, MA) equipped with a HSS-T3 column (100 × 2.1 mm, 1.8 μm particles size; Waters) at a column compartment temperature of 35°C. Mobile phase A is H₂O+0.1%formic acid and mobile phase B is acetonitrile+0.1%formic acid (UHPLC-MS grade solvents, Fisher chemical). Samples were separated with a flow rate of 0.3 mL/min using the following gradient: 1 min isocratic at 5% B, up to 98 B in 16 min, back to 5% B in 0.5 min and then 2.5 min isocratic at 5% B. Q-Exactive HFX mass spectrometer (Thermo Fisher Scientific, Bremen, Germany) was coupled to the UHPLC system. Mass spectra were acquired in both electrospray ionization (ESI) positive and negative modes using data-dependent acquisition (DDA) mode with a mass range of *m/z* 90–1,300. The MS/MS spectra were obtained from the top10 most intense MSI ions. The stepped normalized high-energy dissociation (HCD) collision energies (CEs) of 20,40, and 60 units were used. Capillary temperature is 320°C and probe heater temperature is 350°C. Compound annotation was done by matching accurate mass, isotopic distributions and MS/MS spectral to reference data from an in-house standards TCM database (Shanghai Applied Protein Technology CO., Ltd., Shanghai, China) and public databases GNPS (Wang et al., 2016), ReSpect (Sawada et al., 2012) and Massbank (Horai et al., 2010).

Molecular docking

Molecular docking was performed to evaluate the binding affinity of the 38 identified compounds from SX with PAK1 and RAC1. The 3D structures of PAK1 (PDB ID: 4ZJI) and RAC1 (PDB ID: 3TH5) were retrieved from the RCSB Protein Data Bank. The structural data of the 38 ligand compounds were obtained from the PubChem database. Water molecules, non-essential ligands, and redundant chains were removed from the protein structures using PyMOL. Molecular docking was then carried out using AutoDockTools (v1.5.7) (Eberhardt et al., 2021) with the addition of all hydrogen atoms to prepare the proteins for docking simulations.

Molecular dynamics simulation

The molecular docking conformation with the optimal binding energy was selected for molecular dynamics simulation. Using GROMACS 2023.2 as the computational platform, the system was constructed under the CHARMM36 all-atom force field. The complex was solvated in a cubic water box with a minimum distance of 1.0 nm from the solute to the box boundary, using the TIP3P water model. Na⁺ and Cl[−] ions were added to neutralize the system charge. Following 50,000 steps of energy minimization via the steepest descent algorithm, the system was equilibrated successively in the NVT and NPT ensembles for 1,000,000 steps and 2 ns each, employing the V-rescale thermostat and Berendsen barostat with a time constant of 0.1 ps. Long-range electrostatic interactions were treated with the Particle Mesh Ewald (PME) method, while short-range van der Waals and Coulomb interactions were truncated at 1.0 nm. The Verlet cutoff scheme and LINCS algorithm were applied to constrain hydrogen bonds, allowing a 2 fs integration time step. After equilibration, a production molecular dynamics simulation was performed for 100 ns under conditions of 300 K and 1.0 bar. To

ensure the reliability of the results, each system was simulated in triplicate (Knapp et al., 2018).

In vitro and in vivo experiment

Preparation of the SX aqueous extract

SX is a compound composed of four traditional Chinese medicinal herbs: *Spatholobus suberectus* Dunn (Jixueteng), *Sargentodoxa cuneata* (Oliv.) Rehd. et Wils. (Daxueteng), *Toddalia asiatica* L. Lam. (Jianxuefei), and *Periploca forrestii* Schltr (Heiguteng). The four medicinal herbs were purchased from the Guiyang region of Guizhou Province, China, and were botanically authenticated by Guizhou University of Traditional Chinese Medicine. The aqueous extract of SX was prepared using the method described previously (Wu et al., 2022).

Preparation of herbal serum of SX

Experimental rats received oral administration of SX at a dose of 25.4 g/kg daily, with 12-hour intervals between administrations, continuing for 7 consecutive days. During the final administration, the dosage was doubled. Approximately 1.5–2 hours after the last oral gavage, rats were anesthetized, and blood was collected via the abdominal aorta. The collected blood was allowed to stand at 4°C for 0.5–1 hour, then centrifuged using a cold centrifuge at 4°C, 3,000 rpm for 10 minutes. The supernatant was collected as the SX-containing serum. Prior to use, the serum underwent heat inactivation at 56°C and was subsequently sterilized by filtration through a 0.22 μm membrane filter to ensure purity and eliminate potential contaminants.

Isolation of neutrophils

C57BL/6 mice were euthanized by cervical dislocation and the surface was disinfected with alcohol (Swamydas et al., 2015). The femur and tibia were isolated, and bone marrow cells were flushed out using RPMI 1640 medium. Red blood cells were lysed. The bone marrow cell suspension was carefully layered onto a density gradient formed by reagents A and C and centrifuged at 1000 × *g* for 30 minutes. The lower cell layer was carefully aspirated. Cells were washed by centrifugation at 250 × *g* for 10 minutes. This washing step was repeated twice to obtain purified neutrophils.

Cell culture

Isolated neutrophils were cultured in RPMI 1640 medium (22400089, Gibco, USA) supplemented with 10% fetal bovine serum and 1% penicillin (100 U/ml)–streptomycin (100 μg/ml) solution. Cells were maintained at 37°C in a humidified atmosphere containing 5% CO₂.

Giemsa staining

Neutrophil suspensions were smeared onto slides and air-dried. Cells were fixed with methanol for 5 minutes. After drying, samples were covered with Giemsa stain for 30 seconds, followed by the addition of 2–3 volumes of 1 × PBS for 10 minutes. Slides were rinsed under running tap water to remove excess dye, air-dried, and mounted with resin.

Flow cytometry

Isolated neutrophils were resuspended in FACS buffer (BL1136A, Biosharp, Beijing, China) at a density of 1 × 10⁷/mL and incubated with an anti-CD16 antibody (bs-6028R, Bioss, Beijing, China) for Fc blocking at 4°C in the dark for 15 minutes. The cells were then stained with Anti-mouse-CD45-APC-Cy7 (bsm-30251A), Anti-mouse-CD11b-PE-Cy5 (bs-6028R), and Anti-mouse-Ly6G-FITC (bs-23111R) (all from Bioss, Beijing, China) at 4°C for 30 minutes in the dark (Wishart et al., 2023). Data were acquired using a BD FACS Celesta flow cytometer (BD, USA) and analyzed with FlowJo software.

Cytotoxicity assay

Cytotoxicity of phorbol 12-myristate 13-acetate (PMA), SX, and LGT

was evaluated using the CCK-8 assay. Cells were seeded in 96-well plates at 1×10^5 cells/well and treated with various drug concentrations for specified durations. Then, 20 μ L of CCK-8 solution was added to each well. After 4 hours of incubation, the absorbance was measured at 450 nm using a microplate reader.

Real-time quantitative PCR

Neutrophils (1×10^7 cells/well) were seeded in 6-well plates, stimulated with PMA at different concentrations for 4 hours, and then treated with serum containing SX for 12 hours. Inflammatory and anti-apoptotic factor expressions were quantified using qRT-PCR. Total RNA was extracted using Trizol (Vazyme, China). Reverse transcription was performed according to the Vazyme kit protocol (R233-01). PCR cycling conditions were set using ChamQ SYBR qPCR Master Mix (High ROX Premixed) (Vazyme, China): 2 minutes at 95°C and 50°C, followed by 40 cycles of 15s at 95°C, 2s at 2°C, and 60s at 60°C. Primer sequences are presented in Table 2. Data were analyzed using Bio-Rad CFX Maestro and Excel 2016 software. Relative gene expression was calculated using the comparative Ct method ($2^{-\Delta\Delta C_t}$). Glyceraldehyde 3-phosphate dehydrogenase (GAPDH) served as the reference gene.

Immunofluorescence

Neutrophils were seeded in RPMI 1640 medium containing control or SX-supplemented serum on poly-L-lysine-coated slides in 12-well plates (1×10^6 cells/mL). PMA was added and cells were incubated for 4 hours. Cells were fixed in 4% paraformaldehyde and permeabilized with 0.4% Triton X-100 for 15 minutes. Cytoskeletal staining was performed using Phalloidin-FITC, and nuclei were counterstained with DAPI. Cell analysis and imaging were conducted using a confocal microscope (Olympus, Japan).

Cell migration assay

A Transwell chamber with 3.0 μ m pore size was used, with a 24-well plate as the lower chamber. The lower chamber contained 600 μ L of RPMI 1640 medium supplemented with 10% serum and 0.1 μ M chemotactic factor N-Formyl-Met-Leu-Phe (fMLP). The upper chamber received 200 μ L of cell suspension (1×10^6 cells) and was incubated at 37°C with 5% CO₂ for 30 minutes. Cells in the lower chamber were carefully collected and counted to calculate neutrophil migration rate.

Enzyme-linked immunosorbent assay (ELISA)

IL-1 β , TNF- α , and Caspase-3 levels were measured using ELISA kits. Neutrophils (1×10^7 cells/well) were seeded in 6-well plates, stimulated with PMA (500 ng/mL) for 4 hours, and then treated with serum containing SX for 12 hours. Supernatants were collected for ELISA analysis. The procedure involved sample loading, initial incubation (37°C, 1 h), plate washing, enzyme-conjugated antibody incubation (37°C, 30 min), substrate color development, and reaction termination. Optical density (OD) was measured at 450 nm (reference wavelength 630 nm) using a microplate reader. Concentrations were determined using the standard curve interpolation method with recombinant proteins provided in the kit.

Construction of CIA rat model and drug administration

A total of 48 specific pathogen-free (SPF) male Sprague-Dawley rats, aged 6-8 weeks and weighing 180-200 g, were supplied by the Experimental Animal Center of Guizhou Medical University (license number:

SCXK (Xiang) 2019-0014). The animal study protocol was approved by the Animal Ethics Committee of Guizhou Medical University (approval number: 2200095). The rats were randomly divided into six groups ($N = 8$ per group): control, model, high-dose Sidaxue (SX, 40 g/kg), medium-dose SX (20 g/kg), low-dose SX (10 g/kg), and tripterygium glycosides (GTW, 9 mg/kg) control group (Zhang et al., 2021). Under aseptic conditions, bovine type II collagen (Chondrex, Inc., Redmond, WA, USA) was emulsified with an equal volume of incomplete Freund's adjuvant (IFA) (Chondrex, Inc., Redmond, WA, USA) on ice. After anesthesia, each rat received a subcutaneous injection of 0.2 mL of the emulsion at the base of the tail for initial immunization. A booster injection of 0.1 mL emulsion was administered subcutaneously in the plantar region one week later. Rats in the control group received an equal volume of 0.9% NaCl following the same schedule (Gao et al., 2021; Li et al., 2024).

Arthritis index (AI) scoring and foot swelling assessment

The severity of arthritis in both hind ankle joints was evaluated using a cumulative AI score based on the Table 3. The volume of the left and right hind paws was measured by volumetric displacement. The paw swelling percentage was calculated as follows: Swelling (%) = (Volume after modeling – Volume before modeling) / Volume before modeling $\times$ 100%.

Sample collection

After a 12-hour fast with water withheld following the final administration, the rats were euthanized. The thymus and spleen were collected, and the corresponding organ indices were calculated as the ratio of organ weight to body weight. Hind paw tissues were dissected; one portion was fixed in 4% paraformaldehyde at room temperature for subsequent hematoxylin and eosin (H&E) staining, while the remaining samples were stored at –80°C for Western blot analysis.

Histopathological observation of synovial tissue

Synovial tissues from the knee joints, which had been fixed in 4% paraformaldehyde for 48 hours, were processed through embedding, sectioning, deparaffinization, rehydration, HE staining, dehydration, clearing, and mounting. The pathological changes were observed under an optical microscope.

Western blotting

Neutrophils were seeded in six-well plates at a density of 1×10^7 cells per well and stimulated with PMA at 500 ng/mL for 4 hours. They were subsequently treated for 12 hours with serum containing the SX drug, in the presence or absence of Fingolimod hydrochloride (HY-12005, MedChemExpress, Shanghai, China), followed by protein extraction. Following stimulation, cells were treated with serum containing SX for 12 hours, after which protein extraction was performed. Total protein was extracted from neutrophil or synovial tissue samples using RIPA lysis buffer (Solarbio Science & Technology Co., Ltd., Beijing, China) followed by centrifugation. Protein concentration was determined using a bicinchoninic acid (BCA) assay kit (Solarbio Science & Technology Co., Ltd., Beijing, China). Proteins were separated by sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE) and transferred onto 0.22 μ m polyvinylidene difluoride (PVDF) membranes. The membranes were blocked with 5% bovine serum albumin

Table 2
Primers used for real-time polymerase chain reaction.

Gene	Forward	Reverse
IL-1 β	CTGTTTCTAATGCCTTCCC	TGGTTTCTGTGACCCCTGA
TNF- α	TTGGGATCATTGCCCTGTG	GGCTACAGGCTTGCTACTCG
Caspase 3	TCTGACTGGAAGCCGAAAC	GGCAAGCCATCTCTCATC
GAPDH	GAAGCTGGTCATCAACGGGA	GGCGGAGATGATGACCCTTT

Table 3
The criteria of arthritis index score.

Characteristics	Score
No swelling or erythema observed	0
Slight swelling or mild erythema in one or two joints	1
Mild swelling and erythema affecting multiple joints	2
Moderate swelling and noticeable erythema extending to metatarsal or metacarpal joints	3
Severe swelling and extensive redness throughout the ankle and foot	4

(BSA) (Solarbio Science & Technology Co., Ltd., Beijing, China) for 2 hours at room temperature. Thereafter, the membranes were incubated overnight at 4 °C with the following primary antibodies: β -actin (1:10000, Bioworld, Nanjing, China), GAPDH (1:10000, Bioworld, Nanjing, China), RAC1 (1:10000, Bioworld, Nanjing, China), PAK1 (1:10000, Baijia, Jiangsu, China), p-PAK1 (1:10000, Bioworld, Nanjing, China), LIMK1 (1:10000, Baijia, Jiangsu, China), p-LIMK1 (1:10000, Bioworld, Nanjing, China), Cofilin1 (1:10000, Baijia, Jiangsu, China), p-Cofilin1 (1:10000, Baijia, Jiangsu, China), myeloperoxidase (MPO) (Wanleibio, Shenyang, China), and citrullinated histone H3 (Cit-H3) (1:10000, Abcam, Shanghai, China). After incubation, the membranes were washed six times for 5 minutes each with TBST buffer and then incubated with horseradish peroxidase-conjugated goat anti-rabbit secondary antibody (1:10000, Bioworld, Nanjing, China) at room

temperature for 2 hours with gentle shaking. Following another six washes with TBST, protein bands were visualized using a chemiluminescence substrate, and images were captured with a gel imaging system. Band intensities were quantified using ImageJ software.

Statistical analysis

Statistical analyses were conducted using GraphPad Prism 10.4.1. Continuous variables are presented as mean $\pm$ standard deviation ($\bar{X} \pm s$). The normality of data distribution was assessed using the Shapiro-Wilk test. For comparisons between two groups, if the data followed a normal distribution, an unpaired two-tailed Student's *t*-test was applied; otherwise, the Mann-Whitney *U* test was used. For comparisons among three or more groups, one-way analysis of variance (ANOVA) was

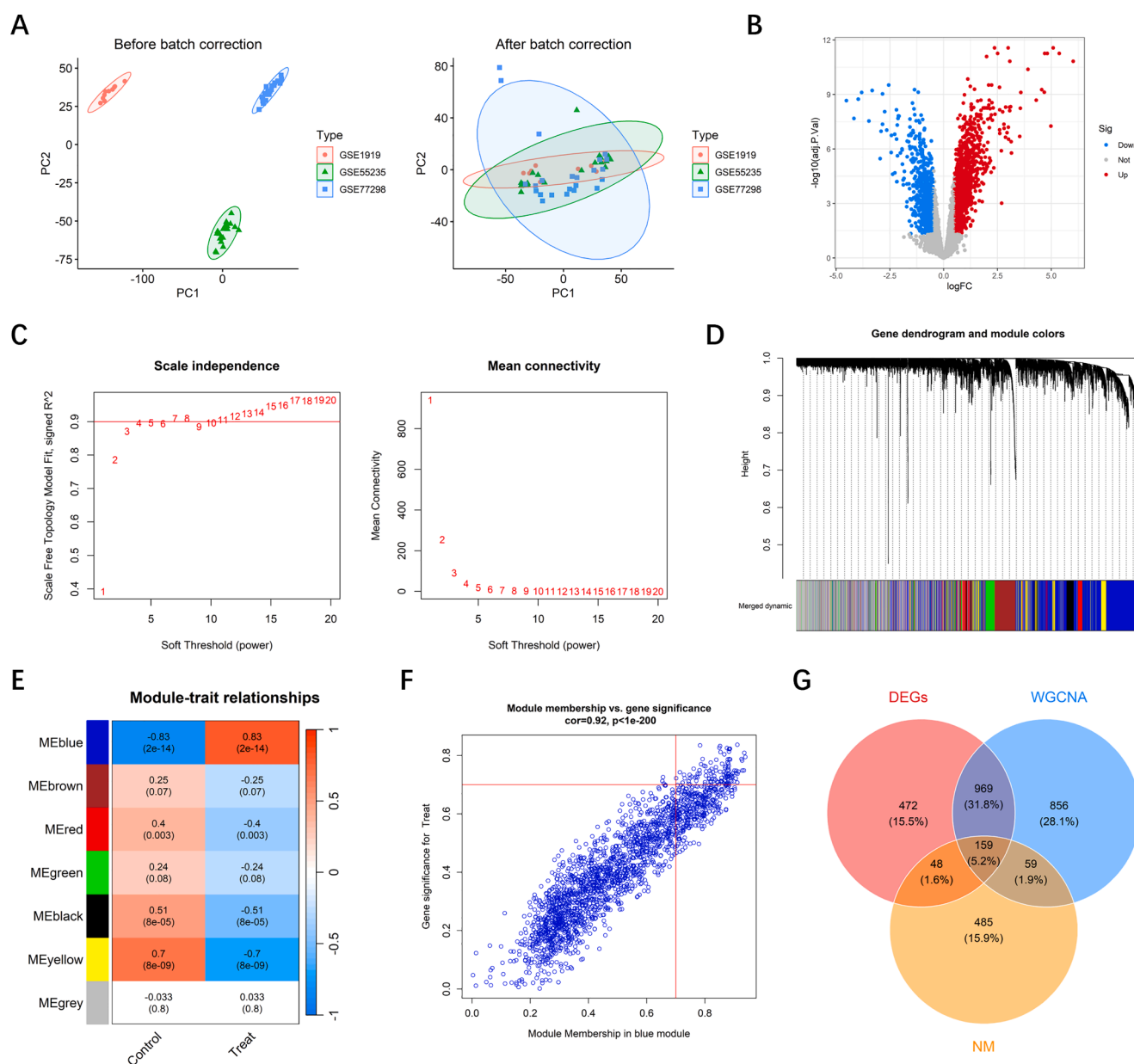


Fig. 1. Screening of DEGs related to neutrophil migration in RA. (A) PCA plot before and after batch effect correction. (B) Volcano plot of DEGs. Blue indicates significantly downregulated genes, red indicates significantly upregulated genes, and grey indicates non-significant genes. (C) Scale independence and mean connectivity in the weighted gene co-expression network. At a soft threshold of 7, the network exhibits high scale independence and mean connectivity close to zero. (D) Different co-expression modules within the weighted gene co-expression network. (E) Heatmap showing the correlation of seven modules with RA case and control groups. (F) Correlation between MM and GS for all genes in the blue module. (G) Venn diagram illustrating the intersection among DEGs, blue module genes, and neutrophil migration-related genes.

employed for normally distributed data, followed by Tukey's post hoc test for multiple comparisons. If the data violated the normality assumption, the Kruskal-Wallis test was used, followed by Dunn's post hoc test. A two-tailed P -value < 0.05 was considered statistically significant.

Results

Identification of DEGs associated with neutrophil migration in RA

We initially employed transcriptomics to identify genes associated with neutrophil migration in RA. After batch effect removal (Fig. 1A), we identified 1,648 DEGs using criteria of $|\log_2FC| > 0.5$ and adjusted $P < 0.05$ (Fig. 1B). Subsequently, we constructed a weighted gene co-expression network using a soft threshold of 7 (Fig. 1C, D). The blue module demonstrated the highest correlation with RA ($R = 0.83$, $P = 2E-14$) (Fig. 1E), with GS and MM showing strong correlation ($R = 0.92$, $P < 1E-200$) (Fig. 1F). We retrieved 751 neutrophil migration-related genes from GeneCards (Table S1) and intersected them with the blue module and DEGs, ultimately identifying 159 genes associated with neutrophil migration in RA (Fig. 1G) (Table S2).

Involvement of RAC/PAK-regulated actin cytoskeleton in neutrophil migration in RA

Neutrophil migration is fundamentally a spatiotemporally regulated process driven by actin filament dynamics, with microtubules and intermediate filaments providing structural support. The cyclic polymerization and depolymerization of the cytoskeleton is closely coupled with

signal transduction pathways, enabling cells to rapidly respond to chemotactic gradients and migrate to sites of inflammation (Belliveau et al., 2023; Sprenkeler et al., 2020; Tur-Gracia and Martinez-Quiles, 2021). GO enrichment analysis of the 159 intersecting genes revealed significant enrichment in terms related to leukocyte migration and chemotaxis (Fig. 2A). KEGG pathway analysis indicated strong enrichment in regulation of actin cytoskeleton (Fig. 2B), and functional interaction analysis suggested crosstalk between regulation of actin cytoskeleton, immune responses, and leukocyte migration (Fig. 2C). Notably, within this pathway, RAC/PAK signaling was significantly activated (Fig. 2D), preliminarily suggesting its involvement in neutrophil migration during RA.

To investigate whether activation of RAC/PAK signaling in RA neutrophils is associated with disease pathology, we performed scRNA-seq analysis. We first constructed a bone marrow single-cell atlas from a CIA DBA/1J mouse model (Fig. 3A). A total of 25,339 cells were annotated into 11 cell types: Neutrophils (NEUT), B cells (BC), Monocytes (MON), Pre-B cells (pre-BC), Macrophages (MAC), Erythroid (ERY), NKT cells (NKT), T cells (TC), Mast cells (MC), Plasma cells (PC), and Megakaryocytes (MEG) (Fig. 3B). Neutrophils were the most abundant cell type (Fig. 3C). Expression of both Rac1 ($P < 0.001$) and Pak1 ($P < 0.0001$) was significantly elevated in neutrophils from CIA mice compared to controls and decreased significantly after treatment ($P < 0.0001$), suggesting that pharmacological intervention targeting RAC1/PAK1 signaling may ameliorate RA (Fig. 3D, E). Similarly, Rac1 and Pak1 exhibited significant co-localization within neutrophils (Fig. 3F), and their expression levels were significantly correlated ($R = 0.34$, $P < 0.001$) (Fig. 3G), indicating that RAC1/PAK1 signaling is likely involved in neutrophil migration.

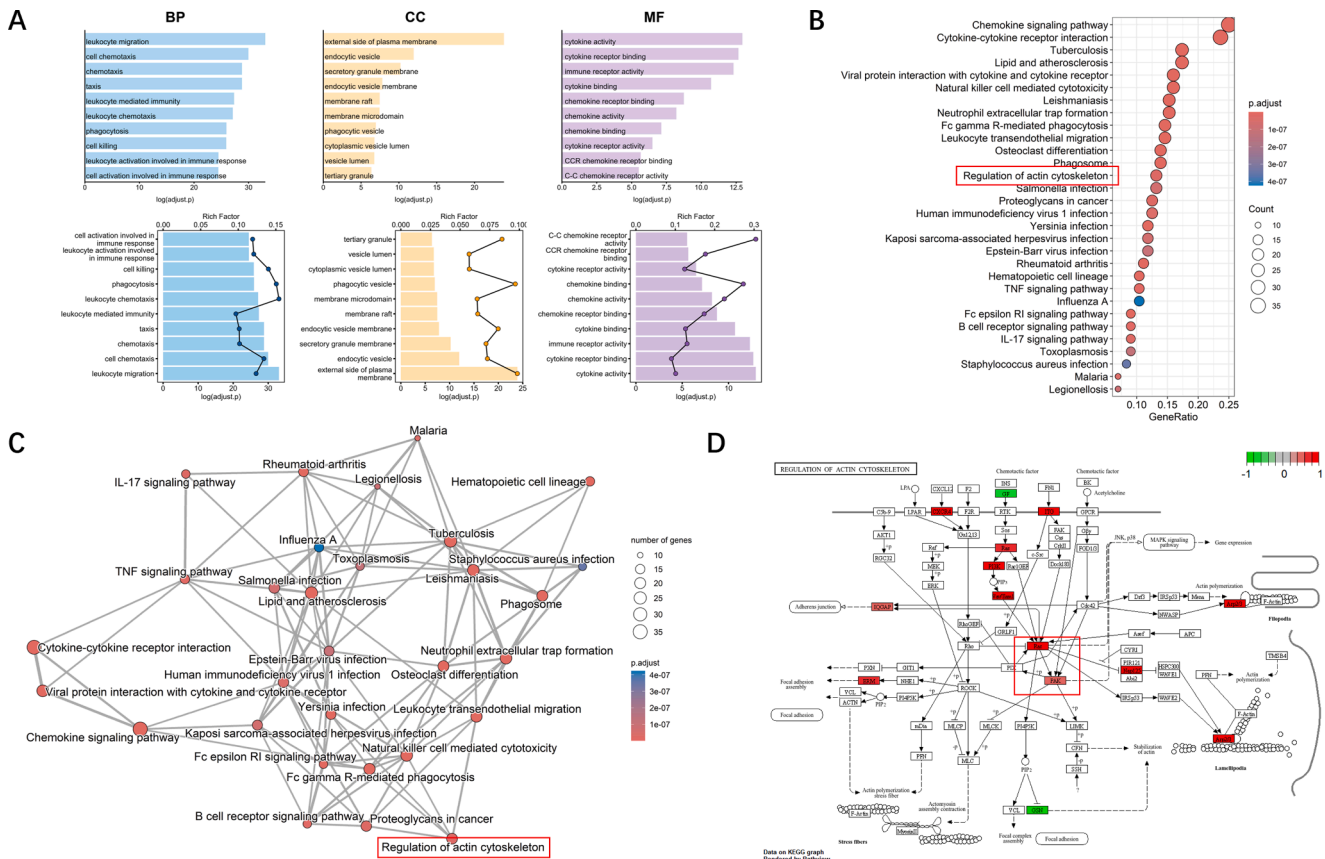


Fig. 2. Functional enrichment analysis of intersecting genes. (A) Results of GO enrichment analysis. Blue represents biological process (BP) terms, yellow represents cellular component (CC) terms, and purple represents molecular function (MF) terms. (B) Bubble chart displaying the results of KEGG analysis. (C) Pathway interaction network of the top 30 enriched KEGG terms. (D) Pathway diagram generated using the pathwayview package, illustrating signal activation within the regulation of actin cytoskeleton.

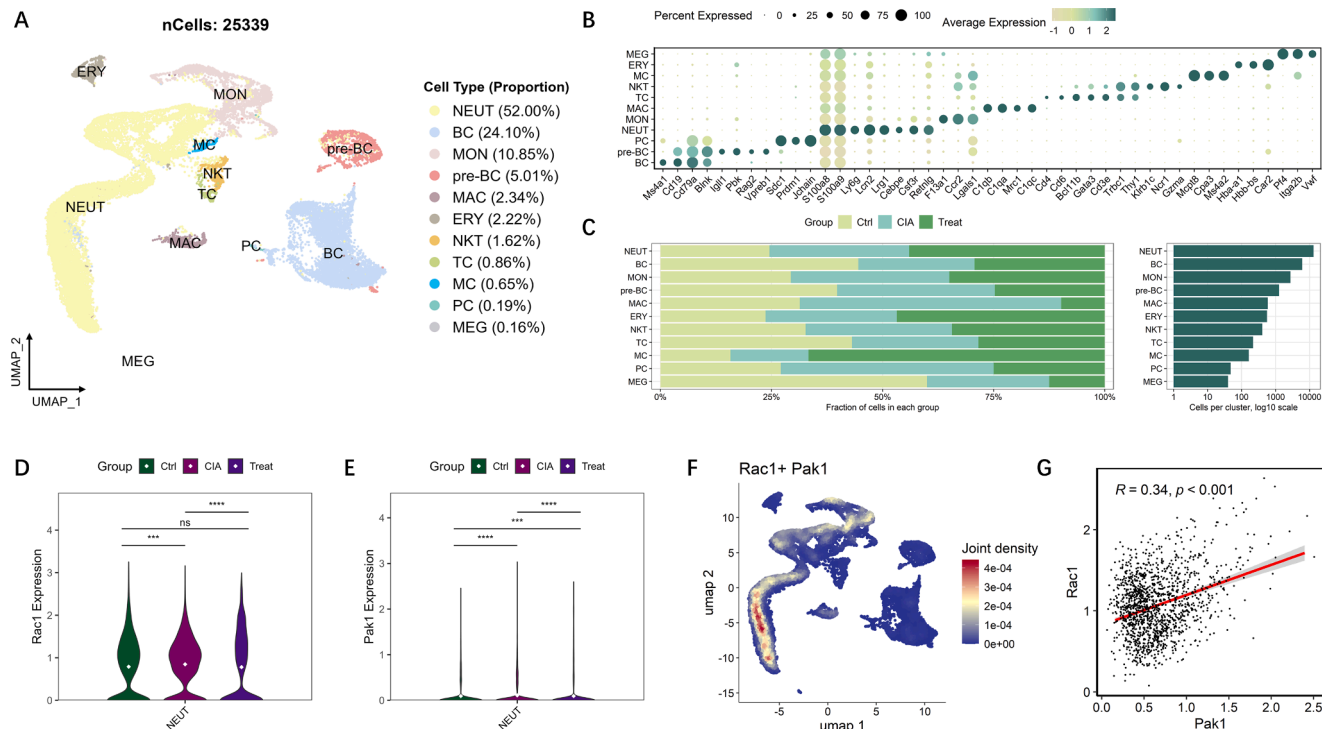


Fig. 3. Results of scRNA-seq Analysis. (A) UMAP projection illustrating the single-cell atlas of bone marrow tissue. (B) Bubble plot showing marker genes for each cell type. (C) Bar plot displaying the proportion and number of cells from each sample source. (D-E) Analysis of Rac1 and Pak1 expression levels in neutrophils across experimental groups. Wilcoxon test with BH-adjusted *P*-values; ns: *P* > 0.05, ***: *P* < 0.001, ****: *P* < 0.0001. (F) UMAP-based heatmap showing co-localization of Rac1 and Pak1 across the atlas; the highest level of co-localization was observed in neutrophils. (G) Correlation between Rac1 and Pak1 expression levels in neutrophils.

Identification of PAK1 as a druggable gene associated with neutrophil migration in RA

PAK1 has previously been identified as a druggable gene in a study by Chris Finan et al. (Finan et al., 2017). To further evaluate whether PAK1 may serve as a druggable target related to neutrophil migration in

RA, we performed MR analysis. The results indicated a significant causal relationship between PAK1 and increased risk of RA (*P* = 0.039, OR = 1.578 (95% CI: 1.024-2.434)) (Fig. 4A–C). Cochran's Q test revealed no evidence of heterogeneity (*P* > 0.05) (Table S3), and the MR-Egger test indicated no significant pleiotropy (*P* > 0.05) (Table S4). Steiger filtering showed no evidence of reverse causality (*P* < 0.05) (Fig. 4D,

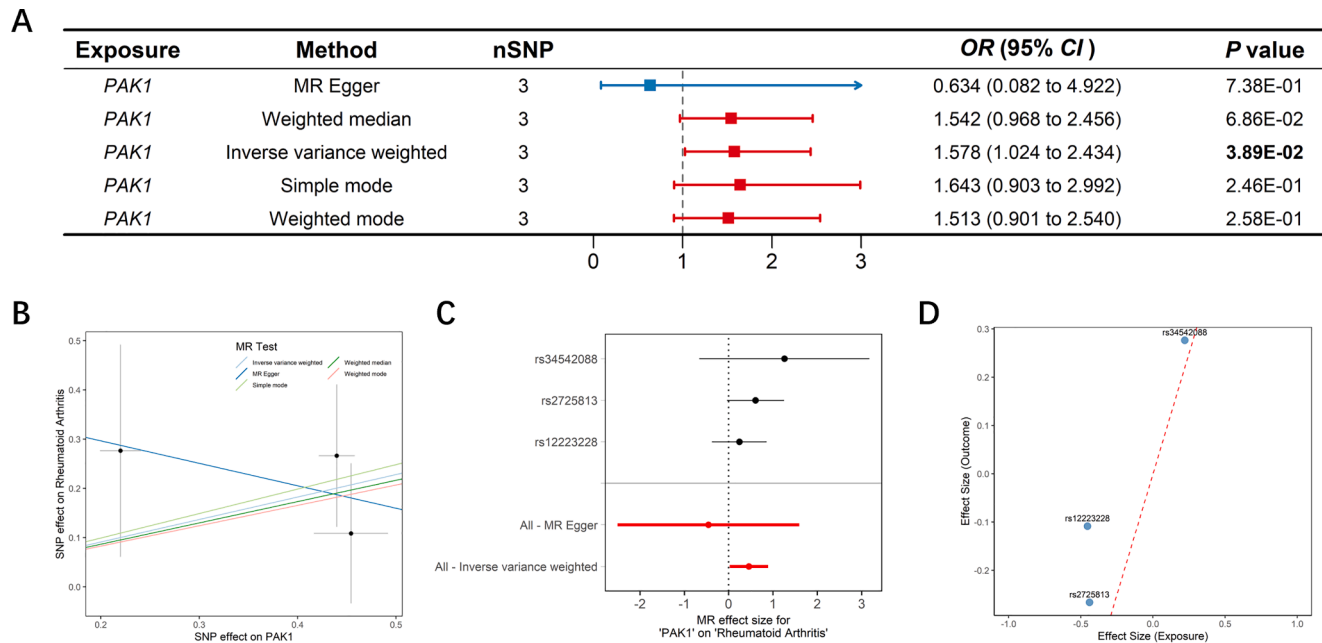


Fig. 4. Results of MR analysis. (A) Forest plot illustrating the causal inference between PAK1 and RA. (B) Radial plot of the MR analysis. (C) Forest plot for each SNP of the MR analysis. (D) Results of Steiger filtering for directionality testing.

Table S5). Therefore, we propose that pharmacological targeting of PAK1 to modulate neutrophil migration may represent a viable strategy for intervening in RA pathogenesis.

Identification of potential therapeutic substances in SX-containing serum

In recent years, reverse network pharmacology approaches have been widely employed to identify potential therapeutics targeting specific molecules, thereby advancing the study of drug mechanisms (Li et al., 2025; Priyamvada et al., 2025; Yin et al., 2022). Our research group has previously demonstrated that SX can ameliorate RA in a CIA rat model, although the underlying mechanism remains unclear (Wu et al., 2022). Therefore, we applied reverse virtual screening to investigate whether SX improves RA by targeting RAC1/PAK1 to modulate neutrophil migration.

In this study, we employed ultra-performance liquid chromatography coupled with high-resolution mass spectrometry to systematically investigate the chemical constituents of SX. Using a non-targeted metabolomics approach, a total of 2,670 compounds were identified by combining both positive and negative ion modes. By comparing with blank serum samples, 766 compounds that potentially enter the systemic circulation were subsequently screened out. From the traditional medicine base peak chromatograms (BPC) in both ionization modes, chromatographic peaks with high abundance were subjected to peak shape confirmation and MS/MS spectrum validation. These peaks were then numbered sequentially, resulting in the annotation of 38 representative compounds (Fig. 5A, B). These components exhibit high signal intensity, well-defined structures, and cover multiple classes such as alkaloids, flavonoids, coumarins, and phenylpropanoids, demonstrating considerable chemical diversity and potential bioactivity (Table 4).

Potential of SX compounds to target RAC1/PAK1

Molecular docking was employed to preliminarily assess the binding affinity between the major compounds of SX and RAC1/PAK1. The results indicated that Mocimycin exhibited the strongest binding to PAK1 (Docking score = -8.99 kcal/mol), while Fumaritine showed the highest affinity for RAC1 (Docking score = -9.54 kcal/mol) (Fig. 6A, B). Notably, all 38 major compounds of SX demonstrated high binding affinity to both RAC1 and PAK1 (Docking score < -5 kcal/mol) (Fig. 6C), suggesting that SX has considerable potential to target RAC1/PAK1 signaling.

Molecular dynamics simulations were performed using the lowest energy docking conformations. The PAK1-Mocimycin system reached equilibrium, with RMSD stabilizing around 0.24 nm (Fig. 6D). The RMSF values were largely below 0.5 nm (Fig. 6E), indicating high stability. The system also exhibited minimal fluctuation in SAS and RG (Fig. 6F, G),

and formed strong hydrogen bonds ranging from 0 to 4 in number (Fig. 6H), confirming a stable binding interaction. Similarly, the RAC1-Fumaritine system also reached equilibrium, with RMSD stabilized at approximately 0.2 nm (Fig. 6I) and all RMSF values below 0.5 nm (Fig. 6J). The SAS and RG values showed almost no fluctuation (Fig. 6K, L), and hydrogen bonds ranged from 0 to 2 (Fig. 6M), indicating a stable complex. In summary, both molecular docking and molecular dynamics simulations provide evidence that the major compounds in SX can effectively bind to RAC1 and PAK1.

Morphological and immunological validation of neutrophils

Neutrophils were isolated from mouse bone marrow and validated morphologically and immunologically. Giemsa staining revealed that the isolated cells exhibited horseshoe-shaped or segmented nuclei (Fig. 7A, B), consistent with neutrophil morphology and possibly reflecting different stages of differentiation. Flow cytometry analysis confirmed that the purity of the isolated neutrophils exceeded 90% (Fig. 7C), indicating that the cells were suitable for subsequent experiments.

SX reduces PMA-induced pro-inflammatory cytokine secretion and neutrophil extracellular trap activation

To determine whether SX-containing serum exerts cytotoxic effects on neutrophil viability, a CCK-8 assay was performed. The results indicated that SX-containing serum at a concentration of 20% exhibited relatively low toxicity toward neutrophils (Fig. 8A). At this concentration, under PMA stimulation, ELISA results showed that PMA stimulation significantly elevated the levels of TNF- α ($P < 0.01$) and IL-1 β ($P < 0.001$), which were notably reversed by 20% SX treatment (Fig. 8B, C). Treatment with SX-containing serum also significantly reduced the mRNA levels of TNF- α ($P < 0.001$), IL-1 β ($P < 0.001$), and Caspase-3 ($P < 0.001$) in neutrophils (Fig. 8D-F). These findings suggest that 20% SX significantly attenuates PMA-induced NETs formation and inflammatory responses. Given the above evidence, 20% SX-containing serum was used in all subsequent in vitro experiments.

To evaluate the potential anti-inflammatory properties of SX, we examined the expression of MPO and Cit-H3, key markers of NETs as well as the levels of inflammatory cytokines. The results demonstrated that PMA stimulation significantly increased the protein expression of MPO ($P < 0.001$) and Cit-H3 ($P < 0.05$), indicating enhanced NETs formation. In contrast, SX treatment markedly reduced the expression of MPO ($P < 0.001$) and Cit-H3 ($P < 0.05$) (Fig. 8G-I). Notably, in the presence of FTY, the inhibitory effects of SX on MPO and Cit-H3 were reversed ($P < 0.001$). This result confirms that SX can inhibit the formation of NETs and suggests that this inhibition may be mediated

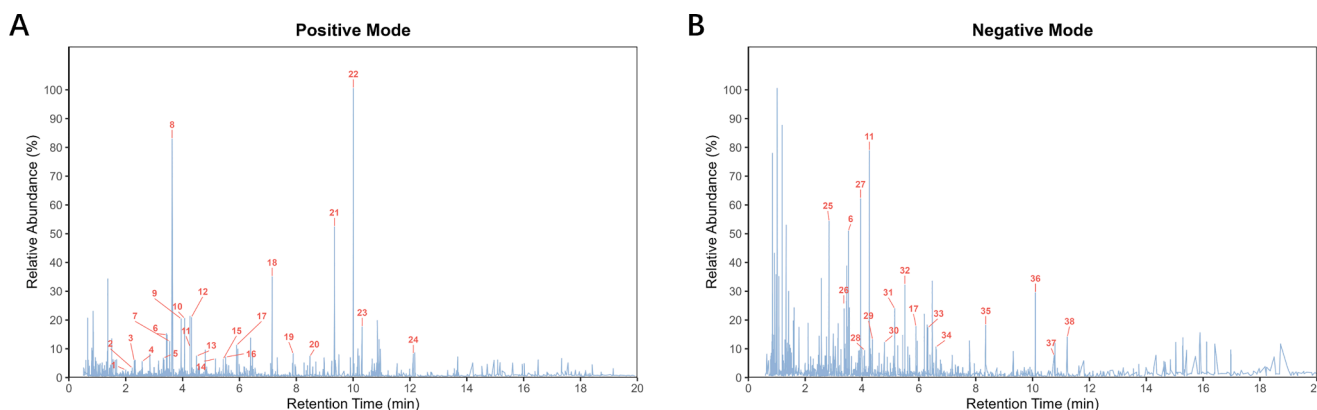


Fig. 5. BPC Analysis of SX. The x-axis represents the retention time range, and the y-axis indicates the mass spectrometric signal intensity. Each numbered label corresponds to an identified peak. (A) Labeled BPC of SX in positive ion mode. (B) Labeled BPC of SX in negative ion mode.

Table 4
Identification of components of SX from labeled BPC peaks.

Numer	Mode	RT min	m/z	ppm	compound name	PubCHEM ID
1	POS	2	178.0859	2	Nigelline	21368
2	POS	2.23	194.1172	2.23	(-)-Salsoline	46695
3	POS	2.3	166.0859	2.3	Valylphenylalanine	144746
4	POS	2.58	307.0806	2.58	(-)-Gallocatechin	9882981
5	POS	3.32	356.1484	3.32	Fumaritine	366269
6	POS	3.46	163.0384	3.46	Heriguard	6537496
7	POS	3.55	579.1483	3.55	Procyanidin B2	11250133
8	POS	3.63	342.1694	3.63	Magnoflorine	73337
9	POS	3.95	291.0853	3.95	Epicatechin	72276
10	POS	4.07	314.1742	4.07	Ethidium	3624
11	POS	4.26	369.1173	4.26	3-O-Feruloylquinic acid	9799386
12	POS	4.31	249.1116	4.31	Glu-Thr	6998031
13	POS	4.49	286.1429	4.49	(S)-Coclaurine	160487
14	POS	5.16	291.0969	5.16	N-Malonyltryptophan	5199636
15	POS	5.43	207.0648	5.43	7-Hydroxy-4-(methoxymethyl)coumarin	5393168
16	POS	5.51	303.0857	5.51	5-Hydroxy-2-(4-hydroxy-3-methoxyphenyl)-4-oxo-3,4-dihydro-2H-chromen-7-yl 6-O-(4-carboxy-3-hydroxy-3-methylbutanoyl)-2-O-(6-deoxy-.alpha.-L-mannopyranosyl)-.beta.-D-glucopyranoside (NP-006983)	45359652
17	POS	5.9	439.1589	5.9	2-(4-Hydroxy-7-methyl-5-oxo-2,3-dihydro-5H-furo[3,2-g]chromen-2-yl)propan-2-yl hexopyranoside (NP-000846)	23899994
18	POS	7.15	309.1324	7.15	Toddalolactone	160485
19	POS	7.89	277.1062	7.89	Visamminol	3083595
20	POS	8.48	260.0911	8.48	Skimmianine	6760
21	POS	9.34	247.0593	9.34	Isopimpinellin	68079
22	POS	10.01	247.0593	10.01	Pimpinellin	4825
23	POS	10.32	233.0439	10.32	Methyl 4-amino-2,3-difluoro-5-nitrobenzoate	2759749
24	POS	12.12	301.1066	12.12	Phellopterin	98608
25	NEG	2.84	353.0872	2.84	Neochlorogenic acid	9998242
26	NEG	3.36	417.1034	3.36	5,7-Dihydroxy-2-(4-hydroxyphenyl)-3-[3,5,7-trihydroxy-2-(4-hydroxyphenyl)chroman-8-yl]chroman-4-one (NP-013445)	45360286
6	NEG	3.52	353.0874	3.52	Heriguard	6537496
27	NEG	3.95	289.0714	3.95	Catechin	9064
28	NEG	4.09	337.0927	4.09	(Z)-2-.beta.-D-Glucopyranosyloxy-4-methoxycinnamic acid (NP-011216)	24094157
11	NEG	4.26	367.1029	4.26	3-O-Feruloylquinic acid	9799386
29	NEG	4.37	581.2232	4.37	(+)-Lyoniresinol 2.alpha.-O-.beta.-D-glucopyranoside (NP-005128)	24121292
30	NEG	4.79	206.0811	4.79	N-Acetyl-L-phenylalanine	74839
31	NEG	5.15	245.0917	5.15	Asn-Asn	16122514
32	NEG	5.51	609.1821	5.51	Hesperidin	10621
17	NEG	5.9	483.1504	5.9	2-(4-Hydroxy-7-methyl-5-oxo-2,3-dihydro-5H-furo[3,2-g]chromen-2-yl)propan-2-yl hexopyranoside	23899994
33	NEG	6.32	187.0965	6.32	Azelaic acid	2266
34	NEG	6.6	137.0233	6.6	Salicylic acid	338
35	NEG	8.35	825.4266	8.35	1-O-((2.beta.,3.beta.,5.Xi.,9.Xi.,18.Xi.)-3-(.beta.-D-Glucopyranuronosyloxy)-2,23-dihydroxy-28-oxoolean-12-en-28-yl)-.beta.-D-glucopyranose (NP-000498)	23900059
36	NEG	10.1	329.2332	10.1	(Z)-5,8,11-trihydroxyoctadec-9-enoic acid	24096399
37	NEG	10.76	795.4162	10.76	Mocimycin	135871572
38	NEG	11.22	663.3744	11.22	Medicagenic acid 3-O-beta-D-glucoside	13732714

through PAK1.

SX inactivates the RAC1/PAK1/LIMK1/cofilin1 signaling pathway and impairs neutrophil migration

Based on our previous findings that RAC1/PAK1 signaling activation regulates cytoskeletal reorganization and promotes neutrophil migration in RA, we further investigated the protective mechanism of SX-containing serum against PMA-induced inflammatory responses in neutrophils. Cytoskeletal changes were visualized using Phalloidin-FITC staining and immunofluorescence microscopy,. The results showed that treatment with 500 ng/mL PMA significantly disrupted the neutrophil cytoskeleton and induced pseudopod extension ($P < 0.01$) (Fig. 9A–C). However, these PMA-induced alterations were reversed after 12 hours of treatment with 20% SX-containing serum ($P < 0.05$) (Fig. 9A–C). Assessment of cell migration using the Transwell assay revealed that PMA treatment significantly enhanced neutrophil migration ($P < 0.05$), which was markedly reduced following SX treatment ($P < 0.001$) (Fig. 9D), indicating that inflammatory stimulation promotes cytoskeletal rearrangement and neutrophil migration, thereby exacerbating inflammatory infiltration.

Notably, PMA treatment increased RAC1 protein levels ($P < 0.001$) and promoted PAK1 phosphorylation ($P < 0.05$), activating RAC1/PAK1

signaling (Fig. 9E–G). In contrast, SX treatment for 12 hours suppressed both RAC1 upregulation ($P < 0.001$) and PAK1 phosphorylation ($P < 0.05$) (Fig. 9E–G). These results suggest that SX attenuates PMA-induced activation of RAC1/PAK1, thereby stabilizing the cytoskeleton, reducing neutrophil migration, mitigating inflammatory infiltration, and exerting anti-inflammatory effects. Since existing studies indicate that RAC1/PAK1 signaling activates downstream LIMK1/Cofilin1 phosphorylation to regulate cytoskeletal dynamics and chemotaxis (Tong et al., 2018; Wufuer et al., 2021), we also examined phosphorylation levels of LIMK1 and Cofilin1. As expected, PMA treatment enhanced the phosphorylation of both LIMK1 ($P < 0.001$) and Cofilin1 ($P < 0.05$), which was significantly reversed by SX treatment (Fig. 9E, H, I). Notably, administration of FTY markedly abolished the effects of SX on Rac1 expression ($P < 0.001$) as well as on the phosphorylation of PAK1 ($P < 0.001$), LIMK1 ($P < 0.001$), and Cofilin1 ($P < 0.05$) (Fig. 9E, H, I). Collectively, these findings demonstrate that SX ameliorates PMA-induced neutrophil migration and inflammatory responses by inhibiting the RAC1/PAK1/LIMK1/Cofilin1 signaling pathway

SX alleviates CIA rat pathology by inhibiting neutrophil migration via blockade of the RAC1/PAK1/LIMK1/Cofilin1 pathway

We established a CIA rat model to further validate *in vivo* whether SX

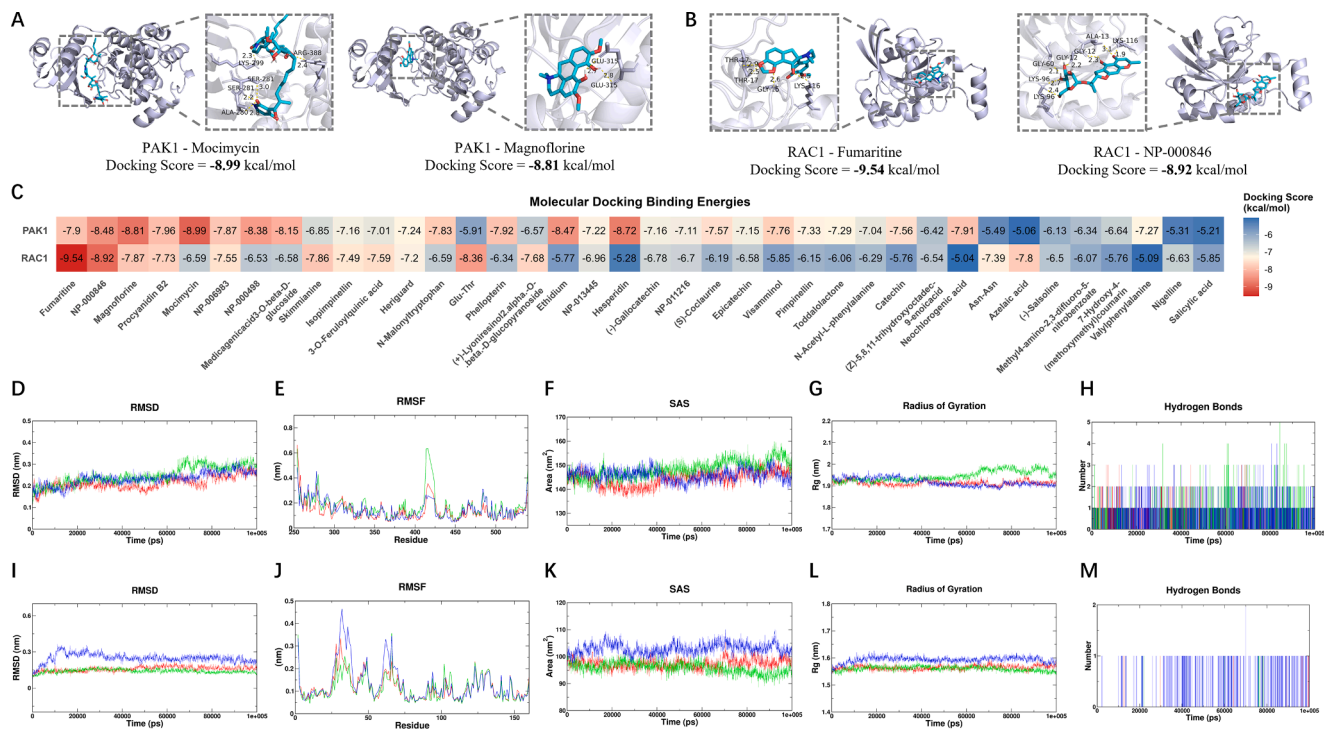


Fig. 6. Molecular Docking and Molecular Dynamics Simulation Results. (A) Visualization of the two compounds with the lowest binding energy to PAK1. (B) Visualization of the two compounds with the lowest binding energy to RAC1. (C) Heatmap of docking scores for 38 SX compounds with PAK1 and RAC1. (D-H) RMSD, RMSF, SAS, RG, and number of hydrogen bonds for the PAK1-Mocimycin system. (I-M) RMSD, RMSF, SAS, RG, and number of hydrogen bonds for the RAC1-Fumaritine system. The red, green, and blue colors represent the first, second, and third replicates, respectively.

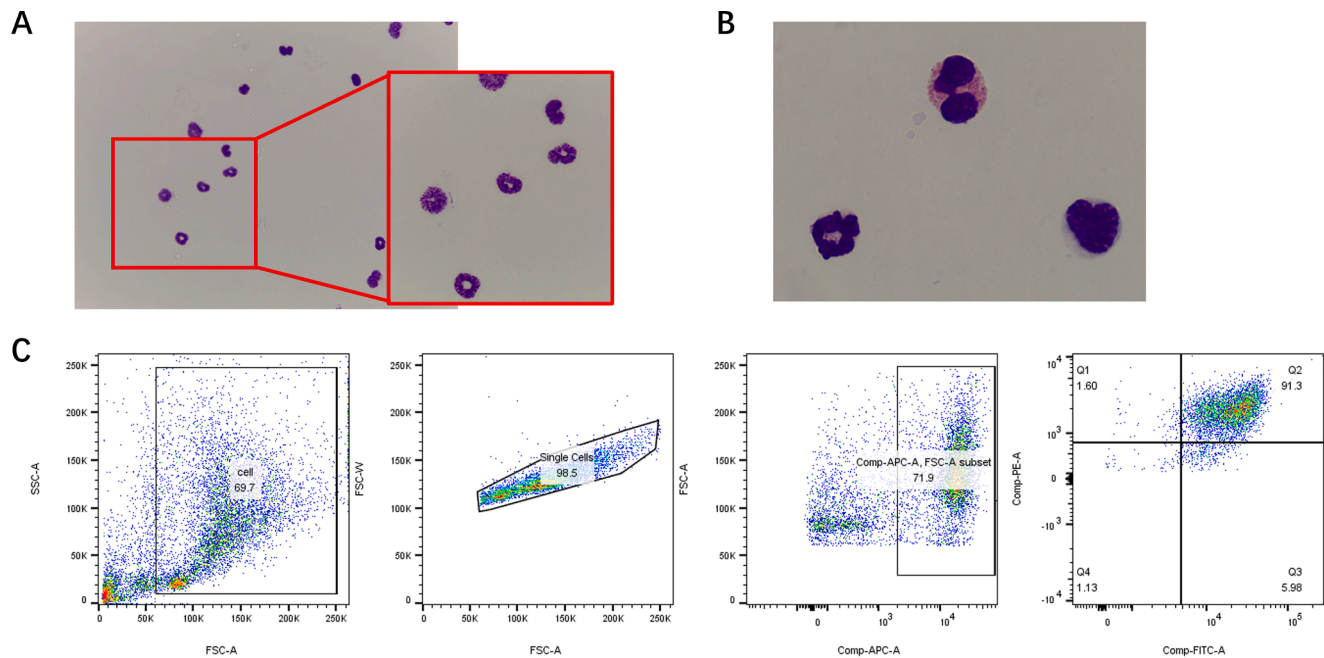


Fig. 7. Morphological and immunological validation of neutrophils. (A-B) Giemsa staining of neutrophils. (C) Flow cytometry validation of neutrophil purity.

inhibits neutrophil migration and ameliorates RA by blocking the RAC1/PAK1/LIMK1/Cofilin1 pathway (Fig. 10A). The results showed that three weeks of high-dose SX treatment significantly reduced paw swelling ($P < 0.01$) (Fig. 10B, C) and AI scores ($P < 0.001$) (Fig. 10D). H&E staining indicated that synovial inflammation was alleviated in the high-dose SX group (Fig. 10E), accompanied by significantly decreased thymic and splenic indices ($P < 0.001$) (Fig. 10G), suggesting that SX

markedly improves RA pathology in rats. Western blot results revealed that the Model group exhibited significantly elevated protein levels of MPO ($P < 0.01$) and RAC1 ($P < 0.01$), as well as increased phosphorylation of PAK1 ($P < 0.01$), LIMK1 ($P < 0.001$), and Cofilin1 ($P < 0.001$) (Fig. 10F, H-L), indicating activation of the RAC1/PAK1/LIMK1/Cofilin1 pathway and enhanced neutrophil migration in the synovium of CIA rats. Consistent with *in vitro*

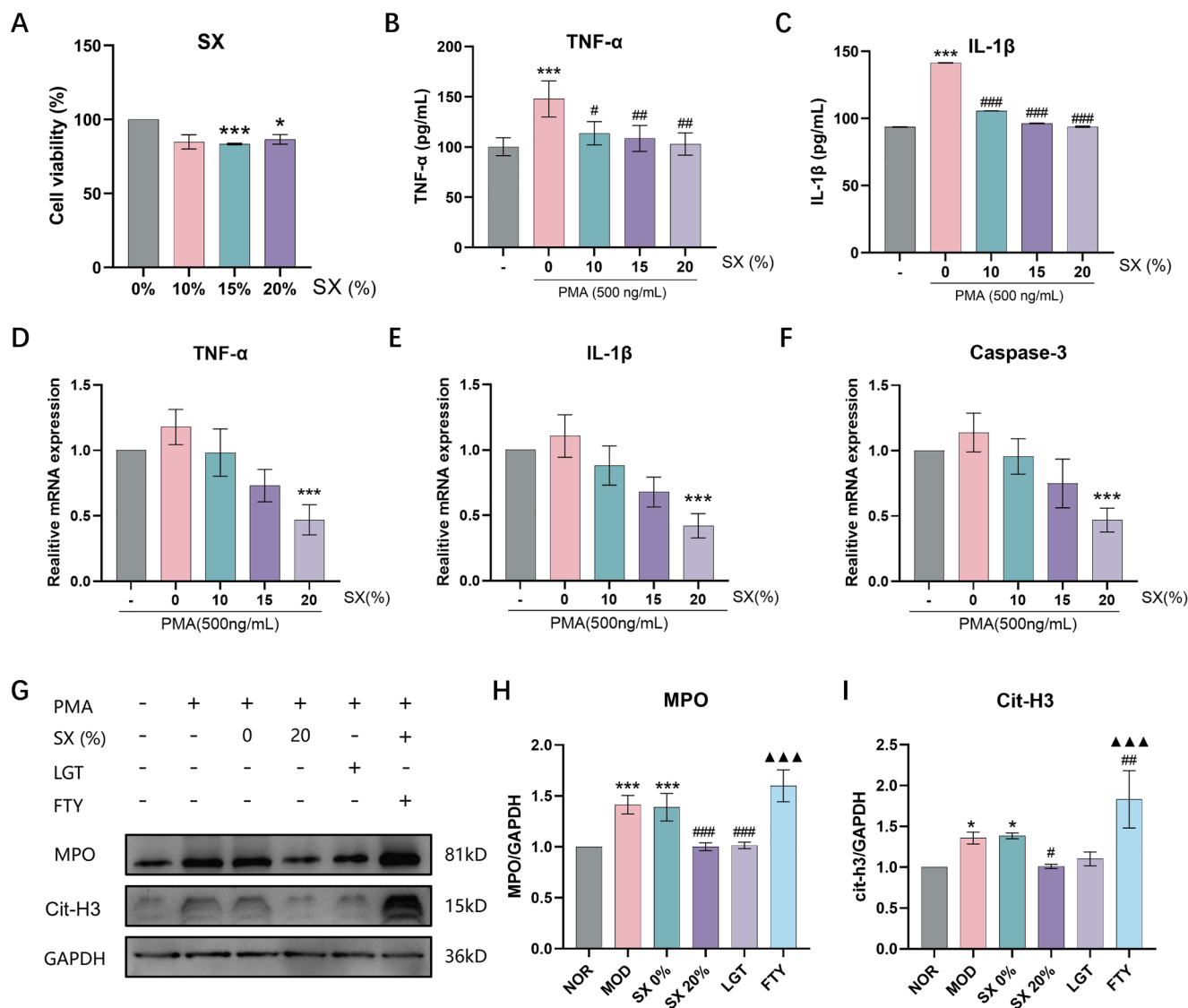


Fig. 8. SX Attenuates PMA-Induced Inflammatory Responses in Neutrophils. (A) Cell viability measured by CCK-8 assay after neutrophils were treated for 12 hours with SX-containing serum at 0%, 10%, 15%, and 20% concentrations. * vs 0% SX group, * $P < 0.05$, *** $P < 0.001$. (B-C) ELISA analysis of cell culture supernatants after 4 hours of PMA stimulation followed by treatment with various concentrations of SX-containing serum. * vs untreated control; # vs 0% SX group; * $P < 0.05$, ** $P < 0.01$; ## $P < 0.05$, ### $P < 0.01$, #### $P < 0.001$. (D-F) mRNA levels of TNF- α , IL-1 β , and Caspase 3 detected by qRT-PCR. * vs untreated control; * $P < 0.05$, ** $P < 0.01$. (G) Representative Western blot images showing protein expression of MPO and Cit-h3. (H-I) Quantitative analysis of protein expression levels in neutrophils after PMA stimulation, with LGT as a positive control. * vs NOR group, # vs Model group, Δ vs SX 20% group; * $P < 0.05$, *** $P < 0.001$; # $P < 0.05$, ### $P < 0.001$; $\Delta\Delta\Delta P < 0.001$.

results, high-dose SX treatment significantly reduced RAC1 protein expression ($P < 0.01$), decreased phosphorylation of PAK1 ($P < 0.01$), LIMK1 ($P < 0.01$), and Cofilin1 ($P < 0.001$), and lowered MPO protein levels ($P < 0.001$) (Fig. 10F, H-L), providing further evidence that SX inhibits neutrophil migration by suppressing the RAC1/PAK1/LIMK1/Cofilin1 pathway.

Discussion

In this study, we employed an integrated analytical approach combining bulk RNA-seq, scRNA-seq, and MR to identify activation of the RAC1/PAK1 signaling pathway during neutrophil migration in RA, highlighting its potential as a druggable target. We then used UHPLC-MS to identify 38 major compound constituents of SX. Through molecular docking and molecular dynamics simulations, we demonstrated that these compounds exhibit strong binding potential to RAC1 and PAK1, suggesting their capability to modulate neutrophil migration via this

pathway. Finally, both *in vitro* and *in vivo* experiments confirmed that SX inhibits neutrophil migration in RA, mechanistically linked to cytoskeletal reorganization regulated by the RAC1/PAK1/LIMK1/Cofilin1 signaling axis. In summary, this multi-omics investigation not only provides novel insights into natural product-based therapeutic strategies for RA but also establishes a theoretical foundation for understanding the pharmacological composition and mechanism of action of SX.

In RA joint tissues, inflammatory cell infiltration is thought to result primarily from transendothelial migration into inflamed tissue rather than from local cellular proliferation (Filippi, 2019). Neutrophils are among the first leukocytes recruited by chemotactic signals and represent one of the major innate immune cell types. Their migration across the vascular endothelium occurs mainly via the paracellular route, which requires the activation of various adhesion molecules and signaling pathways and is highly dependent on cytoskeletal reorganization (Grönloh et al., 2021).

The actin depolymerizing factor/cofilin (ADF/Cofilin) family

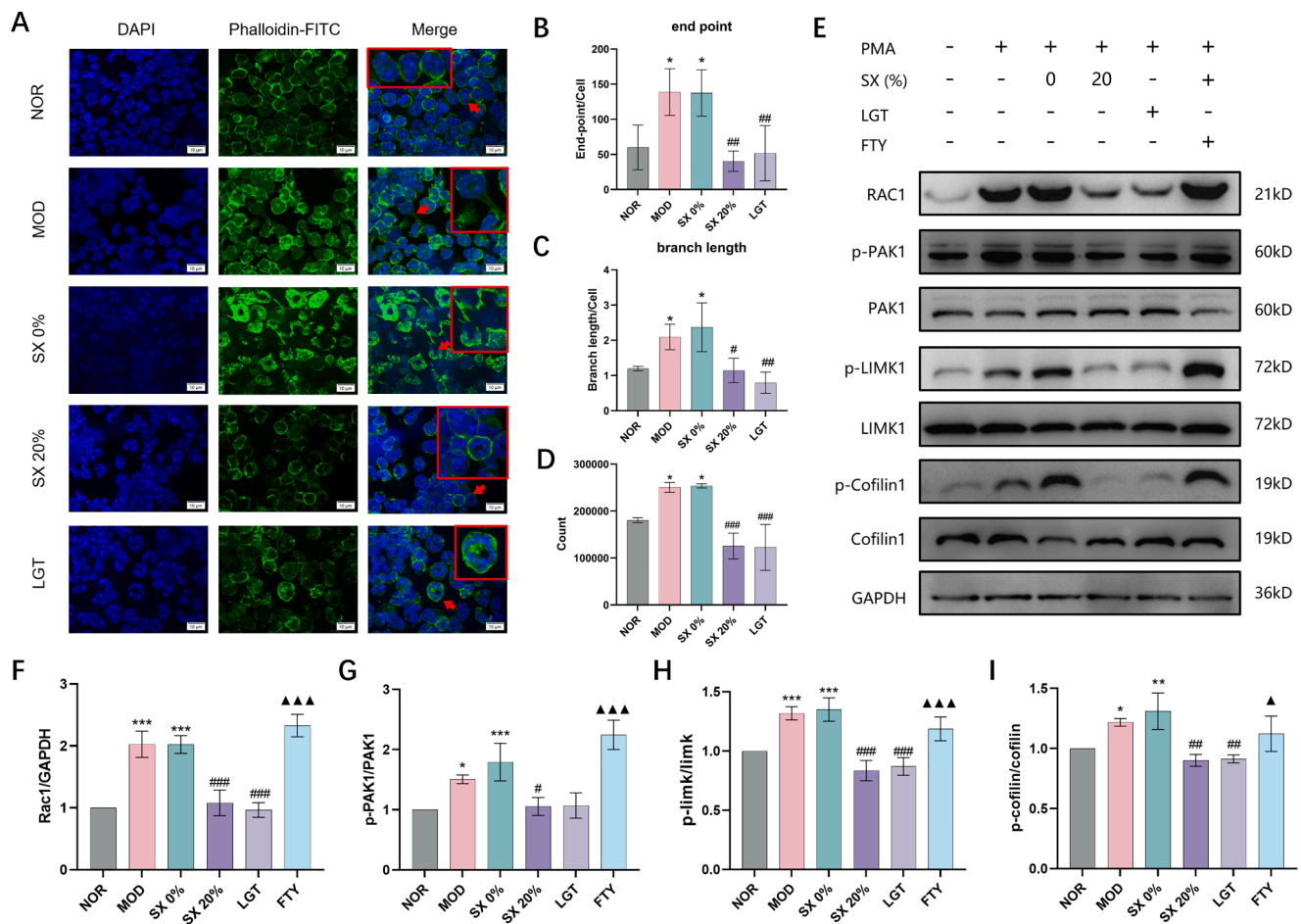


Fig. 9. SX Suppresses Neutrophil Migration via the RAC1/PAK1/LIMK1/Cofilin1 Signaling Pathway. (A) Representative confocal microscopy images showing cytoskeletal changes in neutrophils. The red boxes indicate magnified areas corresponding to the regions marked by the red arrows. (B-C) Statistical analysis results for end point and branch length levels. (D) Cell migration quantified by Transwell assay. (E) Representative Western blot images showing protein expression of RAC1 and phosphorylation level of PAK1/LIMK1/Cofilin1. (F-I) Quantitative analysis of protein expression levels. * vs NOR group, # vs Model group, ▲ vs SX 20% group; * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$; # $P < 0.05$, ## $P < 0.01$, ### $P < 0.001$; ▲ $P < 0.05$, ▲▲ $P < 0.01$, ▲▲▲ $P < 0.001$.

comprises low molecular-weight actin-binding proteins that play a critical role in cell motility (Hamill et al., 2016). Cofilin exists in two states within cells: the active dephosphorylated form and the inactive phosphorylated form (Kapoor, 2014). LIM kinase-1 (LIMK1) phosphorylates cofilin at the third serine residue (Ser3), resulting in its inactivation. This reduces cofilin's affinity for actin, promotes F-actin accumulation, alters cell morphology, and facilitates cell movement. Conversely, dephosphorylation at Ser3 restores cofilin activity, enabling it to sever actin filaments and convert filamentous F-actin into globular G-actin, thereby driving cytoskeletal remodeling (Hu et al., 2019; Niwa et al., 2002; Starinsky-Elbaz et al., 2009). There are two known isoforms of LIMK, LIMK1 and LIMK2. LIMK1 can be activated by multiple upstream signaling pathways, particularly those involving small GTPases. For instance, Rho and Rac GTPases activate downstream effectors such as p21-activated kinase 1 or 4 (PAK1 or PAK4) and Rho-associated protein kinase (ROCK). These kinases phosphorylate threonine residues within the activation loop of LIMK1, thereby modulating its activity (Romarowski et al., 2015; Steinbacher and Ebnet, 2018). Thus, RAC1 activates PAK1, which in turn phosphorylates LIMK1. This leads to the reversible phosphorylation of cofilin, regulating the polymerization of F-actin and influencing cytoskeletal dynamics.

Our study identified PAK1 as a druggable target in RA through an integrated approach combining Bulk RNA-seq, scRNA-seq and MR analyses. PAK1 (p21-activated kinase 1), a serine/threonine protein kinase, plays a crucial role in cytoskeleton remodeling, cell cycle and

proliferation, apoptosis, and signal transduction (Mirzaiebadizi et al., 2025). There is a close association between PAK1 and neutrophils. The phosphorylation of PAK1 is related to the migration of neutrophils. A study by Ken Inoue et al. (Inoue et al., 2017) found that Carbon monoxide-releasing molecules-401 (CORMs) can inhibit the migration ability of neutrophils, and this inhibitory effect is related to the phosphorylation level of PAK1. Specifically, CORM-401 can inhibit fMLP-induced PAK1 phosphorylation, thereby affecting the migratory directionality and speed of neutrophils. Moreover, some studies have also shown that the activation of PAK1 is associated with the activation of NADPH oxidase in neutrophils (Tsai et al., 2012, 2013). Consistent with these findings, we observed that PAK1 phosphorylation activates the LIMK1/Cofilin pathway, promoting neutrophil migration and exacerbating inflammation in RA.

The RAC1/PAK1/LIMK1/Cofilin1 signaling pathway is closely associated with cell migration and has been identified as a therapeutic target for inhibiting cell migration in numerous diseases. For instance, Reziwan Wufuer et al. (Wufuer et al., 2021) reported that Ce6-PDT inhibits the migration of SW620 cells by downregulating the RAC1/PAK1/LIMK1/Cofilin1 signaling pathway, with enhanced inhibitory effects observed upon knockdown of RAC1 expression. Similarly, Shan Zhang et al. (Zhang et al., 2020) found that DEPDC1B promotes PC cell migration and invasion by activating the RAC1/PAK1/LIMK1/Cofilin1 signaling pathway. The same is true for neutrophil migration. Jiangwei Ni et al. (Ni et al., 2023) found that esculin significantly inhibits RAC1

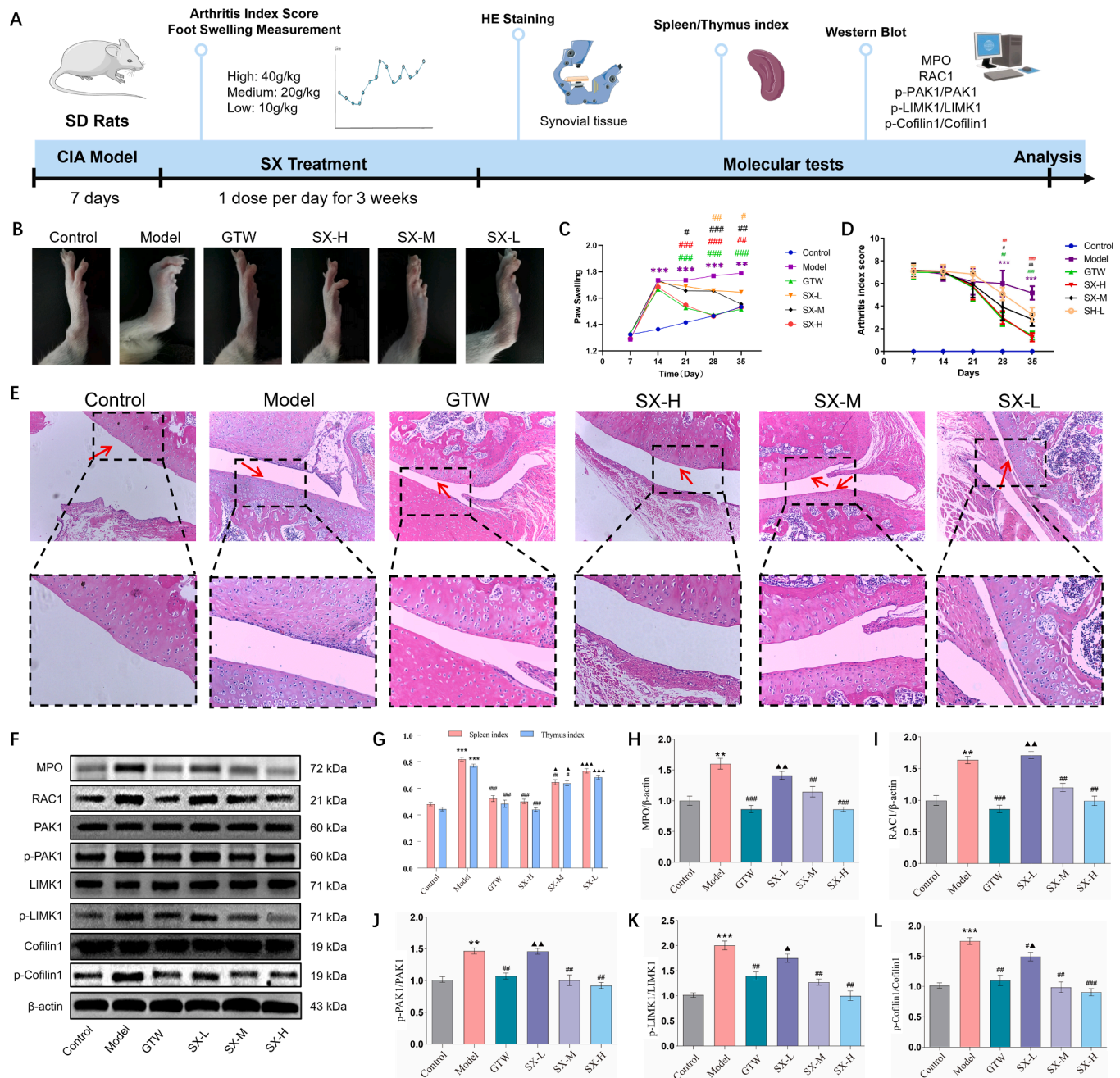


Fig. 10. *In vivo* Experimental Validation. (A) Schematic diagram of the animal experimental procedure. (B-C) Representative images and quantitative results of foot swelling measurements in rats from each group. (D) Changes in the AI scores of rats over three weeks. *** $P < 0.001$ vs Control group; # $P < 0.05$, ## $P < 0.01$, ### $P < 0.001$ vs Model group. (E) H&E staining results of synovial tissues from each group, comprising both panoramic and detailed views. (F) Representative images from Western blot analysis. (G) Spleen and thymus indices of rats in each group. (H) Original band results from Western blot analysis. (I-L) Quantitative results of MPO protein content, RAC1 protein content, p-PAK1/PAK1 ratio, p-LIMK1/LIMK1 ratio, and p-Cofilin1/Cofilin1 ratio. ** $P < 0.01$, *** $P < 0.001$ vs Control group; # $P < 0.05$, ## $P < 0.01$, ### $P < 0.001$ vs Model group; $\Delta P < 0.05$, $\Delta\Delta P < 0.01$ vs GTW group.

activation and the PAK1/LIMK1/Cofilin1 signaling axis, blocking the cytoskeleton remodeling process required for neutrophil recruitment. These studies highlight the strong association between the RAC1/PAK1/LIMK1/Cofilin1 pathway and cell migration, suggesting its potential as a therapeutic target for improving disease outcomes. Similar to these studies, our research demonstrates that the RAC1/PAK1/LIMK1/Cofilin1 signaling pathway regulates neutrophil migration in RA. We further found that SX inhibits this pathway, leading to reduced neutrophil migration and ameliorated RA symptoms.

Despite significant advancements in the treatment of RA over the past few decades, current therapeutic strategies still face numerous

challenges, including issues related to drug accessibility, suboptimal responses in some patients, and limited treatment options for extra-articular manifestations (Uke et al., 2025). Therefore, the development of novel therapeutic strategies remains a crucial direction for future RA research and clinical practice. Existing studies indicate that SX exerts anti-arthritis effects by inducing apoptosis of synovial fibroblasts, downregulating matrix metalloproteinases, and improving synovial angiogenesis (Li et al., 2024; Wang et al., 2022; Wu et al., 2022, 2021). In this study, we reveal that SX reduces neutrophil migration by inhibiting the RAC1/PAK1/LIMK1/Cofilin1 pathway, thereby attenuating RA inflammation. These findings provide further immunological insight

into the mechanism through which SX ameliorates RA.

Notably, we employed UHPLC-MS to help characterize the chemical composition of SX. Major compounds such as Gallocatechin, which has known anti-inflammatory and antioxidant properties (de Oliveira et al., 2014), have been shown to exhibit anti-RA effects *in vitro* (Misra et al., 2022). Procyanidin B2 has been reported to block the infiltration of inflammatory immune cells, including neutrophils, thereby alleviating gout *in vivo* (Qiao et al., 2020). Additionally, Magnoflorine—another principal compound—has been demonstrated to improve RA *in vivo* by suppressing the PI3K/Akt/NF- κ B and MAPK signaling pathways (Shen et al., 2022; Wang et al., 2023). Other components such as Skimmianine (Wang et al., 2010), Isopimpinellin (Robertson et al., 2016), and Azelaic acid (Akamatsu et al., 1991) are also known to inhibit neutrophil migration. Consistent with these reports, our study confirms that SX reduces neutrophil migration and alleviates RA inflammation. Interestingly, the UHPLC-MS analysis also identified several compounds whose potential anti-RA effects have not yet been reported. Given the marked anti-inflammatory and anti-RA activities of SX, extracting its active compounds offers considerable promise for the development of natural-based drugs aimed at treating inflammation and RA.

It is important to note, however, that this study has several limitations. First, the binding affinities between SX-derived compounds and PAK1/RAC1 were predicted through molecular docking and dynamics simulations, which provide supportive but not conclusive evidence of direct target engagement. Future studies employing techniques such as surface plasmon resonance, cellular thermal shift assay, or drug affinity responsive target stability could further validate direct compound-target interactions. Second, as SX is a multi-component herbal formula, its overall pharmacological effect likely arises from the synergistic or additive actions of multiple constituents. The specific contributions of individual compounds within SX to the observed anti-migratory and anti-inflammatory effects remain to be elucidated. We plan to adopt a precision medicine strategy in future research by isolating and studying single compounds derived from SX, thereby clarifying their respective roles and potential synergistic mechanisms. Lastly, while rescue experiments with Fingolimod hydrochloride supported the involvement of PAK1, we did not use genetic knockdown or overexpression techniques to block the RAC1/PAK1/LIMK1/Cofilin1 pathway, which would provide more definitive mechanistic insight into how SX inhibits neutrophil migration. These limitations highlight important avenues for further investigation while underscoring the complexity of studying traditional multi-herb formulations.

Conclusion

In this study, we identified 159 genes associated with neutrophil migration in RA synovial tissue using differential gene expression analysis and WGCNA. Functional enrichment and scRNA-seq analyses revealed that RAC1/PAK1 activation is involved in neutrophil migration

within RA synovium. MR analysis further indicated PAK1 as a druggable target for RA. Through UHPLC-MS analysis, we identified 38 compound constituents of SX, and molecular docking and dynamics simulations confirmed the binding affinity of these compounds to RAC1/PAK1. Cellular experiments demonstrated that SX-containing serum inhibits neutrophil migration, NETs formation, and activation of the RAC1/PAK1/LIMK1/Cofilin1 signaling pathway. Moreover, *in vivo* experiments confirmed that SX suppresses neutrophil migration in the synovial tissue of CIA rats by blocking the RAC1/PAK1/LIMK1/Cofilin1 pathway, thereby ameliorating synovial inflammation (Fig. 11). Based on an immunotherapeutic perspective, this study provides a theoretical foundation for the anti-inflammatory effects of SX and its mechanism of action in improving RA, offering new insights into the development of natural products for RA treatment. Future research may focus on screening specific anti-inflammatory monomers derived from SX.

Ethics approval and consent to participate

All data sources used are publicly available, which have been ethically approved. All animal protocols were approved by the Animal Ethics Committee of the Experimental Animal Center of Guizhou Medical University (approval number: 2200095).

Consent for publication

Not applicable.

Availability of data and materials

All relevant publicly available data are provided within the manuscript. The raw data supporting the findings of this study are available from the corresponding author upon reasonable request.

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Consent to participate

Not applicable.

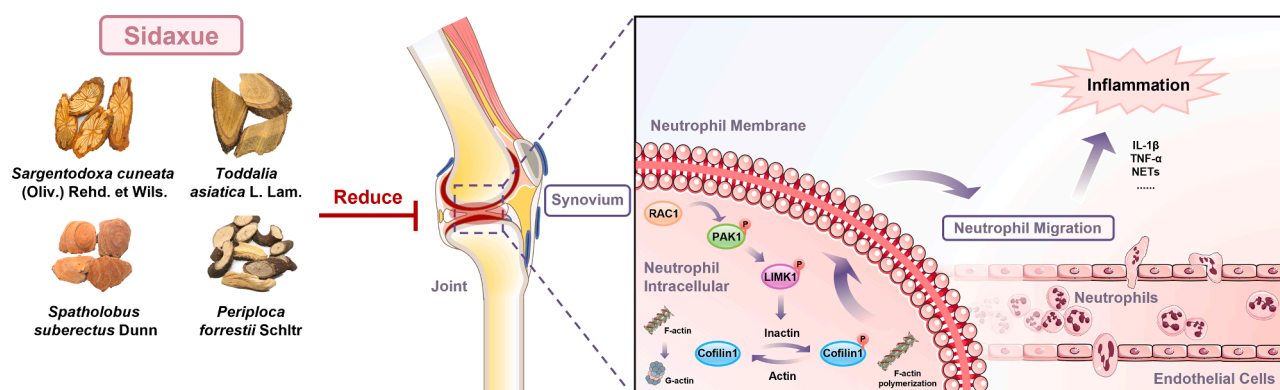


Fig. 11. Schematic diagram of the molecular mechanism by which SX inhibits neutrophil migration to ameliorate RA.

Author agreement

We the undersigned declare that this manuscript entitled “*Sidaxue Ameliorates Rheumatoid Arthritis by Inhibiting Synovial Neutrophil Migration: Insights from an Integrated Multi-omics Analysis*” is original, has not been published before and is not currently being considered for publication elsewhere.

We confirm that the manuscript has been read and approved by all named authors and that there are not other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

We understand that the corresponding author is the sole contact for the Editorial process. He is responsible for communicating with the other authors about progress, submissions of revisions and final approval of proofs.

CRediT authorship contribution statement

Bingyang Xu: Writing – review & editing, Writing – original draft, Validation, Methodology, Data curation. **Shuhan Liu:** Writing – original draft, Validation, Software, Investigation. **Mingxie Chen:** Writing – original draft, Resources, Project administration, Conceptualization. **Xinyue Li:** Writing – original draft, Methodology, Data curation. **Xuan Bi:** Writing – original draft, Visualization, Formal analysis. **Xiangqian Xue:** Writing – original draft, Software, Project administration. **Peijun Cheng:** Writing – original draft, Data curation. **Xiaona Guo:** Writing – review & editing, Writing – original draft. **Leilei Liu:** Writing – original draft, Methodology. **Yingting Yang:** Writing – original draft, Conceptualization. **Jianfei Sun:** Writing – review & editing, Writing – original draft, Supervision, Resources. **Ning Wu:** Writing – review & editing, Writing – original draft, Validation, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.phymed.2026.158050](https://doi.org/10.1016/j.phymed.2026.158050).

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